

Vamsi Makke

vamsimakke.jobs@gmail.com | +1 (513) 601-9627 | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer with **3+ years of experience** building production backend systems and **ML pipelines**. Built **GenAI features** on **AWS Bedrock**, multimodal video classification and embedding-based retrieval, and own the full lifecycle (design, testing, deployment, monitoring) of distributed services for a platform serving **15,000+ MAU**. Strong in **Java, Python, TypeScript**, data structures, and algorithms (Top **4%** on LeetCode, **1,500+** problems solved). MS in IT, **GPA 4.0/4.0**.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, Bash

GenAI & ML: AWS Bedrock (Claude), Embeddings, Vector Search (Pinecone), Semantic Retrieval

Backend: Spring Boot, FastAPI, Node.js, REST APIs, Microservices, WebSockets, Apache Kafka

Frontend: React, TypeScript, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Elasticsearch

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, Bedrock, CloudWatch), Docker, Kubernetes, GitHub Actions, CI/CD

Testing: JUnit, Mockito, Jest, Unit & Integration Testing

Tools: Git, Docker, Postman, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer (Founding Backend Engineer) | **TORQ Sports**, Cincinnati, OH (Remote) Jun 2025 – Present

- Built the entire backend from scratch pre-launch and now operate it at **15,000+ MAU: 4 microservices** across **Node.js/Python FastAPI** and **3 PostgreSQL databases (80+ tables)**, owning the complete lifecycle from design to on-call monitoring.
- Built a **multimodal GenAI pipeline (AWS Bedrock, Claude Haiku)** analyzing **500+ videos/day**: developed FFmpeg frame sampling and vision-language classification to output schema-validated JSON metadata.
- Designed the core feed ranking algorithm, combining time-decay scoring, AI-derived content quality, and engagement signals with incremental re-ranking every 15 minutes.
- Built modular **React + TypeScript** web client and admin dashboards (content moderation, user management, analytics) with reusable typed components, covered by **Jest** unit tests in CI.
- Built an end-to-end video pipeline: synchronous MP4 remux with thumbnails, then asynchronous **HLS** transcoding to 4 quality levels for adaptive bitrate streaming.
- Cut **p95 API latency from 1s to 300ms** through query optimization, composite indexing, and **Redis** caching, root-causing bottlenecks via **CloudWatch** metrics and Postgres slow-query logs.
- Built **CI/CD (GitHub Actions, Docker)** for zero-downtime deployments and **CloudWatch** observability with custom metrics and alarms, cutting incident detection from 30 min to under 5 min.
- Secured the API with **JWT** auth, role-based access control, input validation, and parameterized queries to prevent injection.

Software Engineer | **Cognizant Technology Solutions**, Hyderabad, India Jan 2022 – Dec 2023

- Developed full-stack features for an enterprise resource management platform using **Java, Spring Boot**, and a **React.js** frontend on **PostgreSQL**, serving **1,000+ daily active users** across 5 business units.
- Increased unit and integration test coverage from 40% to **85%** using **JUnit** and **Mockito**, sustained by requiring tests on every pull request in code review, reducing production defects by **60%**.
- Cut report generation from **47s to 8s**: benchmarked queries, used execution-plan analysis to eliminate full-table scans, added composite indexes, and refactored complex joins.
- Implemented a **Redis** caching layer for frequently accessed data, reducing database load by **30%**.
- Built resource-booking REST APIs with **Spring Security** role-based access control, input validation, and handling for concurrent-booking conflicts and cross-region timezones.
- Led a containerization initiative using **Docker**, improving release frequency from bi-weekly to daily deployments.

EDUCATION

University of Cincinnati, Cincinnati, OH Jan 2024 – May 2025

Master of Science in Information Technology | **GPA: 4.0 / 4.0**

Vasireddy Venkatadri Institute of Technology, India

Sep 2019 – Apr 2023

Bachelor of Technology in Computer Science and Engineering

PROJECTS

Real-Time Anomaly Detection Engine | *Python, FastAPI, AWS Bedrock, Pinecone*

- Built an **ML pipeline** for log analysis using **AWS Bedrock** embeddings and **Pinecone** vector search across **1M+ log entries**, flagging anomalies by nearest-neighbor distance to catch semantically similar errors.

- Designed an ingestion pipeline with batched embedding generation (100 logs/batch) to optimize Bedrock API costs while maintaining near-real-time indexing.
- Achieved **50ms p95 latency** for nearest-neighbor retrieval through Pinecone's ANN indexing, reducing false positive alerts by **60%** compared to rule-based approaches.

Distributed E-Wallet System | *Java, Spring Boot, Apache Kafka, Redis, Docker*

- Architected microservices-based digital wallet (User, Wallet, Transaction, Notification) with **Apache Kafka** for event-driven communication to preserve immutable transaction audit trails.
- Implemented the **Saga pattern** with choreography-based coordination for distributed transactions, enabling automatic roll-back across services when any step fails.
- Built **Redis distributed locking** (Redlock) to prevent double-spend, validated with **500 concurrent transactions** achieving zero data inconsistency.

ACHIEVEMENTS

- **LeetCode:** Top **4%** globally with contest rating **1898**, solved **1,500+** problems in data structures and algorithms.
- **Hackathon:** 3rd place at Virtusa Neural Hack S6 National Hackathon among **10,000+** participants.